FNLP
Lanuage Modelling

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Outline

1. **What are Language Models?**
   - Why do we need language models?
2. **N-gram Language Models**
   - Markov Assumptions
   - N-gram Language Models
   - Parameter Estimations
   - Evaluation
3. **Deal with Unseen Events**
   - Linear Interpolation
   - Smoothing
4. **Beyond N-gram Language Models**

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  - the dog laughs
  - dog the dog bark a laugh smile

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  - syntax: grammars, syntactic analysis?
  - pragmatics?...

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  - pragmatics?...
  - **how about relying on data solely?**

Formally (sort of)

- Given a finite size of vocabulary:

$$\mathcal{V} = \{\text{a, an, dog, cat, bite, bark, ... zoo, ...}\}$$

- we can have an infinite set of strings of words, $\mathcal{S}$:
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  - the dog barks . $\rightarrow$ good
  - the dog laughs . $\rightarrow$ good
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  - ...

From a Probabilistic View

There is a probability distribution p over $\mathcal{S}$:

$$\sum_{s \in \mathcal{S}} p(s) = 1, p(s) \geq 0, \text{for all } s \in \mathcal{S}$$

that is :

- p(the dog barks .)
- p(the dog laughs .)
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- p(the dog laughs .) $\rightarrow$ large
- p(dog the bark a laugh smile .) $\rightarrow$ very small
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p: possibly what we want!

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Why Language Model ?

- Speech Recognition
- Optical Character Recognition (OCR)
- Handwriting Recognition
- Word Segmentation (*whatdoesthismean*?)
- Machine Translation
- Language Generation
- Question Answering/Dialogue/Chat

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- recognize speech
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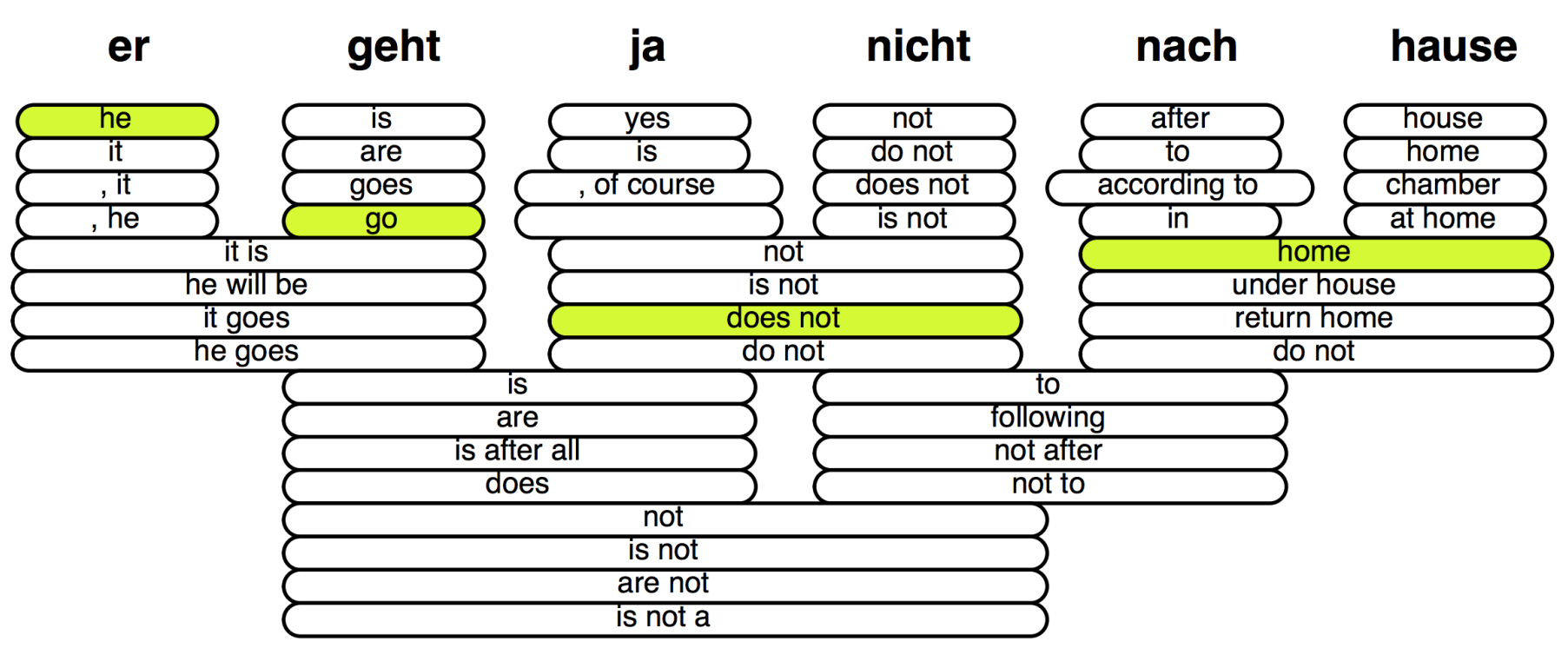
$$words^* = \arg\max_{words} \text{Faithfulness}(signals, words) + \text{Fluency}(words)$$

What can LM do?

Statistical Machine Translation



[from Philipp Koehn]

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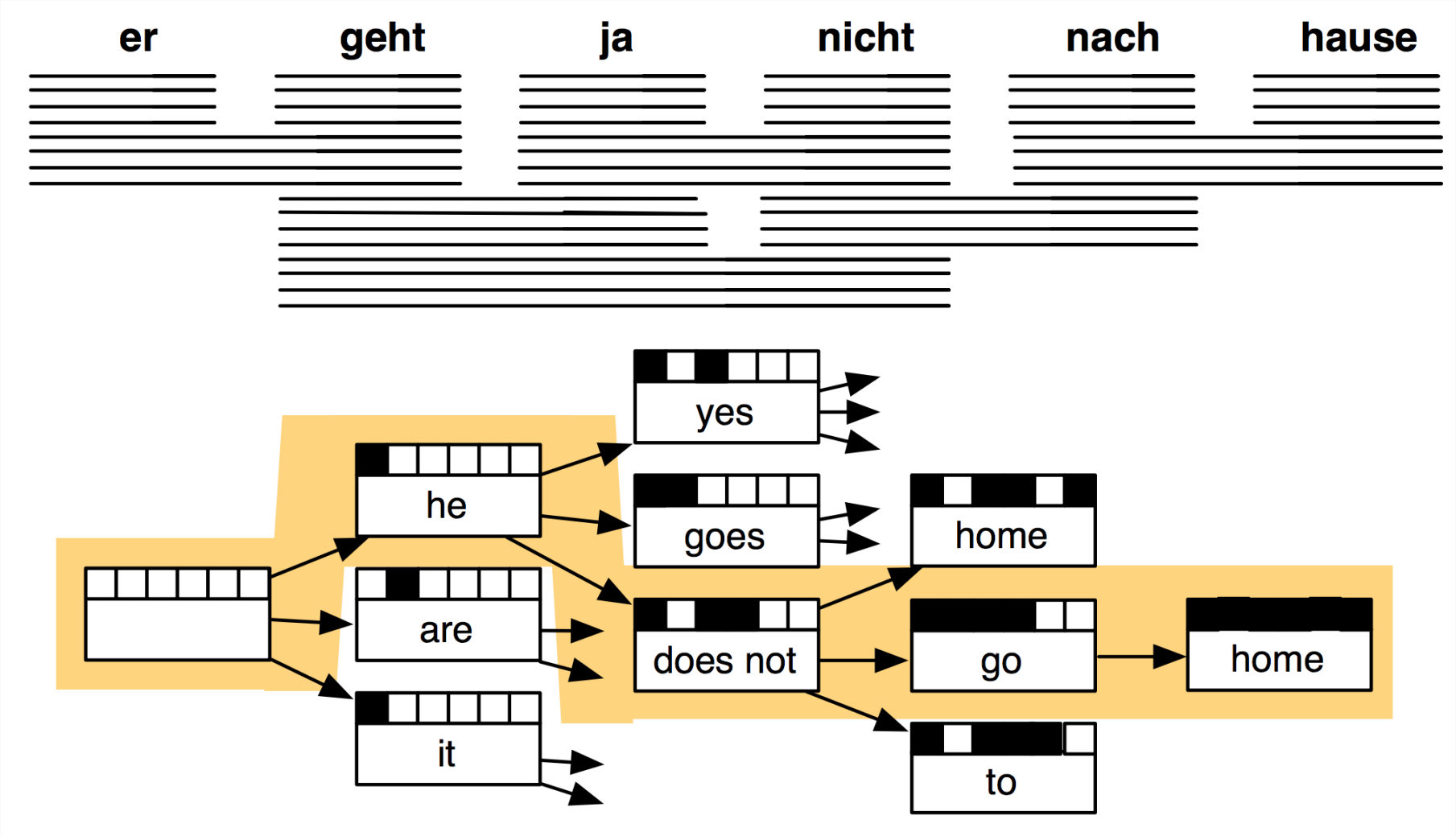

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Text Generation

today the price of gold per ton , while production of shoe lasts and shoe industry , the bank intervened just after it considered and rejected an imf demand to rebuild depleted european stocks , sept 30 end primary 76 cts a share .

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[from Chris Manning]

- Looks good?
- Locally good?
- Globally not...

What can LM do?

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[from Chris Manning]
But, from ChatGPT *help me write a short passage that begins with "today the price of gold per ton"*

Today the price of gold per ton is on the rise, with experts predicting a surge in demand for the precious metal. As the global economy continues to recover from the pandemic, investors are turning to gold as a safe haven asset to protect their wealth from market fluctuations. This has led to a spike in gold prices, with many analysts projecting that the trend will continue in the coming months. Despite the volatility in the market, gold remains a reliable store of value and a popular choice among investors looking to diversify their portfolios.

What can LM do?

Again, for all the above scenarios:

$$words^* = \arg\max_{words} \text{Faithfulness}(signals, words) + \text{Fluency}(words)$$

What can LM do?

Sentence Completion [MS SCC]

- That is his _ fault, but on the whole he's a good worker.
  - generous
  - mother's
  - successful
  - favorite
  - main
- Was she his _, his friend, or his mistress?
  - client
  - musings
  - discomfiture
  - choice
  - opportunity
- All red-headed men who are above the age of _ years, are eligible.
  - seven
  - twenty-one
  - 1,200
  - 60,000
  - 800

语言模型考察的是语言习惯

What can LM do?

How surprise you see the next/masked word given the context?

$$\text{Surprise}(context, answer_1) < \text{Surprise}(context, answer_2)$$

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- *the difficulty of a word is proportional to its surprisal (its negative log-probability) in the context within which it appears.*

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- the so-called **surprisal theory** (Hale, 2001; Levy, 2008)
- *words are easier to comprehend in contexts where they are highly predictable than in unconstraining contexts*

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Intuitively

If we treat every possible sentence s as a single point in a huge space $\mathcal{S}$,

- there are $|\mathcal{S}|$ sentences in $\mathcal{S}$:
- $$p(s) = \frac{count(s)}{|\mathcal{S}|}$$

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  - how big is the $\mathcal{S}$?
  - s can be in various forms
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- Break down? 分解

Another Point of View

- Sentence as a sequence of words $w_1, w_2, w_3, ..., w_n$
- Sentence as a sequence of random variables $X_1, X_2, X_3, ..., X_n$
- The model : $P(X_1 = w_1, X_2 = w_2, X_3 = w_3, ..., X_n = STOP)$
  - This is a **sequence**, not a set! 序列性质，而不是集合
  - $P(X_1 = \text{I}, X_2 = \text{love})$ and $P(X_1 = \text{love}, X_2 = \text{I})$ are different
  - STOP and, sometimes, START 引入标记

A bit Probabilistic

$$P(X_1 = w_1, X_2 = w_2, X_3 = w_3, ..., X_n = STOP)$$
$$= P(X_1 = w_1) \prod_{i=2}^{n} P(X_i = w_i | X_1 = w_1, ..., X_{i-1} = w_{i-1})$$

- Chain Rule

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- when n is big, ...

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- when n is big, ...
- Let's make ASSUMPTIONS! 变小历史的窗口

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- Chain Rule
- when n is big, ...
- Let's make ASSUMPTIONS! $\Rightarrow$ First-order Markov assumption:

马尔科夫一阶独立性假设，只看前面一个词

$$P(X_i = w_i | X_1 = w_1, ..., X_{i-1} = w_{i-1}) \approx P(X_i = w_i | X_{i-1} = w_{i-1})$$

Make Assumption

Let's make stronger ASSUMPTIONS! $\Rightarrow$ no history at all: 0阶

$$P(X_i = w_i | X_1 = w_1, ..., X_{i-1} = w_{i-1}) \approx P(X_i = w_i)$$

- Unigram Language Model 就是朴素贝叶斯那一套
- under MLE estimations:

$$P(X_i = w_i) = \frac{count(w_i)}{\sum_{w_*} count(w_*)}$$

- seems weird, but rather simple, easy to understand
- **cheap**!

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- This is actually bag-of-words (BoW), definitely not good enough

Markov Assumption

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Markov Assumption

- Widely used in many applications!

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Bigram and Trigram LMs

- Widely used in many applications!
- No history assumption → Unigram LM
- First-order Markov assumption → Bigram LM
- Second-order Markov assumption → Trigram LM
- 4-gram LM, 5-gram LM, ...

最高7：存储代价和近似估计代价高

Trigram LM

- A finite vocabulary $\mathcal{V}$
- Parameters $p(a|b,c)$, where b, c, a is an arbitrary trigram $a \in \mathcal{V} \cup \{STOP\}$ and $b, c \in \mathcal{V}$
- For a new sentence $w_1, w_2, ...w_n$

$$p(w_1, w_2, ...w_n) = p(w_1)p(w_2|w_1)\prod_{i=3}^{n} p(w_i|w_{i-2}, w_{i-1})$$

A Toy Example

For the sentence:

the cat laughs . STOP

given a trigram LM, we have:

$$
\begin{aligned}
p(\text{the cat laughs . STOP}) = & p(\text{the}) \\
& \times p(\text{cat}|\text{the}) \\
& \times p(\text{laughs}|\text{the, cat}) \\
& \times p(.|\text{cat, laughs}) \\
& \times p(\text{STOP}|\text{laughs, .})
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$$

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How to Estimate Parameters from Corpora

The **CORE** is:

- $p(w_i|w_{i-1})$ for bigram LM
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核心是怎么估计概率

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and,

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- Maximum likelihood estimation
- More data lead to better model ?

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- 275 million words

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unigram大小和语料库大小一样

- the unigram LM needs to store 716,706 probabilities (at most)
- bigram LM : 12,537,755
- trigram LM : 22,174,483
- ...

说明稀疏性很强

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- 12,537,755 v.s. 51,505,109,010,436 ($\sim$ 1/4,000,000)
- Data sparsity: most are zeros
- Trigram LMs are often good enough

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How do We Evaluate a Language Model

Intuition: a good LM model should assign a real sentence of that language a high probability.

准则：

给常见、好的句子一个好的分数（高的概率）

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- and for the whole dataset, $\prod_{s \in \mathcal{D}} p(s)$
- normalize over all M words in $\mathcal{D}$, $\sqrt[M]{\prod_{s \in \mathcal{D}} p(s)}$

测试集所有句子得分的乘积

比较不同测试集→归一化（实际有多少个词，而不是词汇表大小）

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- simplify with $\log$, $\frac{1}{M} \sum_{s \in \mathcal{D}} \log p(s)$
- also the cross entropy: $-\frac{1}{M} \sum_{s \in \mathcal{D}} \log p(s)$

How do We Evaluate a Language Model

Intuition: a good LM model should assign a real sentence of that language a high probability.

$\rightarrow$ the best LM should best predict an unseen test set.

For a test set $\mathcal{D}$, we can use a language model to

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- now **the perplexity of** $\mathcal{D}$: 混淆度/歧义度

$$Perplexity(\mathcal{D}) = 2^{-\frac{1}{M} \sum_{s \in \mathcal{D}} \log p(s)}$$

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得分差不多→模型不好→歧义度很大

- the lower perplexity, the better LM

Perplexity

For a dataset $\mathcal{D}$, with vocabulary $\mathcal{V}$,

- if for any trigram, we have $p(a|b,c) = \frac{1}{|\mathcal{V}|+1}$ (a uniform distribution over $\mathcal{V} \cup STOP$) for all $b, c \in \mathcal{V} \cup \{*\}$
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- what is the perplexity of $\mathcal{D}$??
- Perplexity $=|\mathcal{V}| + 1$

一个理想的trigram，训练集和测试集分布相同→模型不会犯错→所有句子的概率都趋于1→算出来歧义度是1

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- Perplexity is a measure of *branching factor*
- Considered as the *effective size of the vocabulary*
- Also, to evaluate how hard an NLP task is?

Evaluating Language Models

- use the `Training dataset` to estimate the n-grams, possibly tuned
- compute perplexity on the `Test dataset`

However,

- Better to use test sets that are widely accepted by the community.
- **Perplexity** scores should be compared against the same $\mathcal{V}$.
- Perplexity is NOT a universal metric for downstream applications.

Unigram v.s. Bigram v.s. Trigram ...

Intuitively, which one is better?

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Intuitively, which one is better?

- if n is smaller, ...
  - Does your model learn about the language?
- if n is bigger, ...
  - The cost!
  - The engineering thing! (zeros)
  - ...

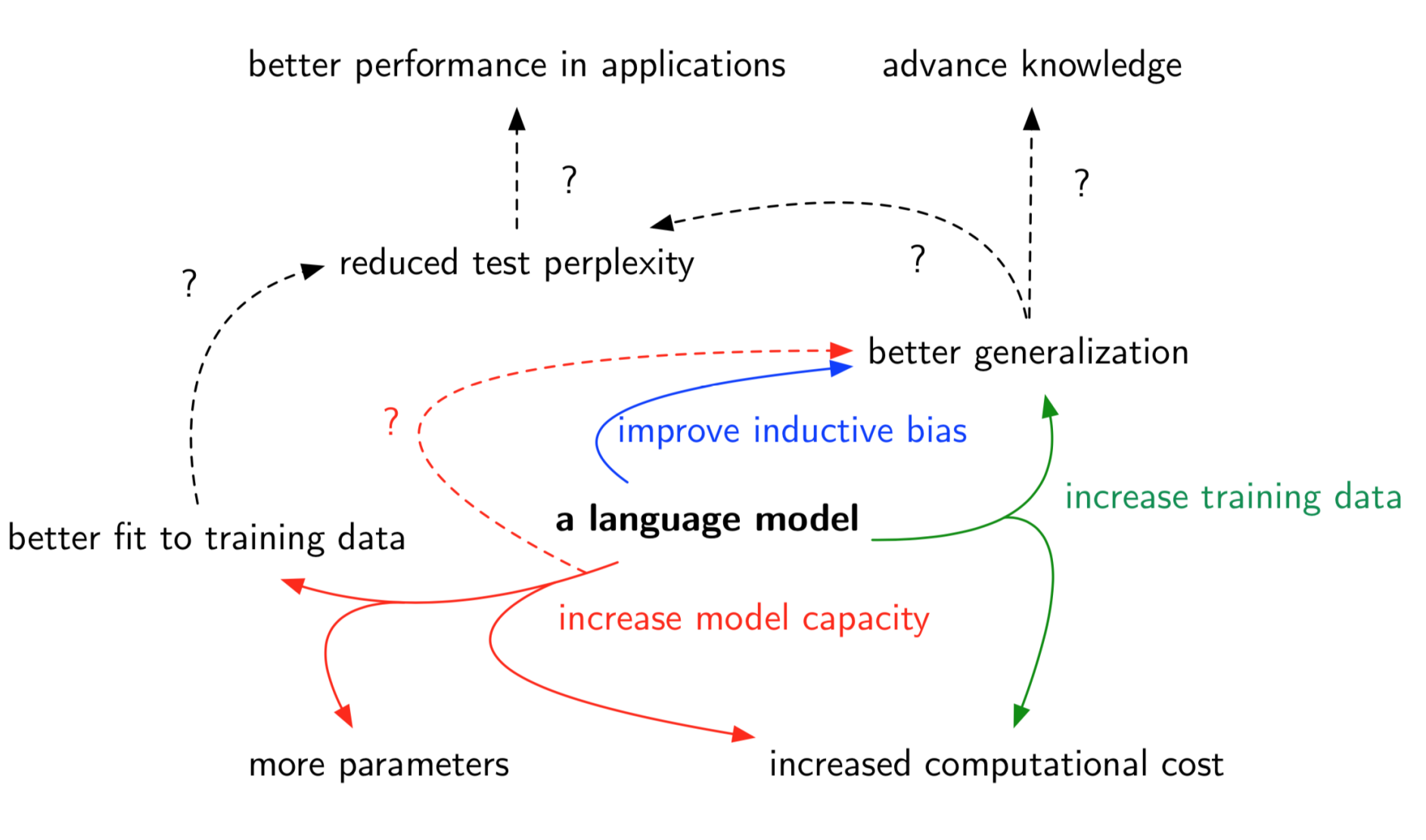
Unigram v.s. Bigram v.s. Trigram ...

Intuitively, which one is better?

- if n is smaller, ...
  - Does your model learn about the language?
- if n is bigger, ...
  - The cost!
  - The engineering thing! (zeros)
  - ...
- if you have more training data, ...
  - perform better
  - generalize better
  - take a look at Google's N-gram

两个方面：增加训练数据/参数（层、计算量）

Language Modelling Research



[from Noah Smith]

An Example

Training set

- i love pku .
- i like thu .
- you love pku .
- you do not like thu .

- Can you estimate a bigram LM from the training data?
  - $p(\text{thu}|\text{like})$
  - $p(\text{like}|\text{i})$

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Test set

- you like pku .
- i hate thu .

- Can you compute the perplexity using the bigram LM ?
  - $p(\text{you, like, pku, ..., STOP})$
  - $p(\text{i, hate, thu, ..., STOP})$

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Unseen Events

- New words
  - *hate*
- New n-grams
  - *you like*, *like pku*

Unseen Events

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- New n-grams
  - *you like*, *like pku*

Recall the MLE estimation ...

Unseen Events

MLE yields bad estimations for Unseen Events

- many Zeros

Unseen Events

MLE yields bad estimations for Unseen Events

- many Zeros

Brain Storm

- What can we do?

Unseen Events

MLE yields bad estimations for Unseen Events

- many Zeros

Brain Storm

We should give some probability mass to those unseen events! This will inevitably reduce the mass for seen events.

Unseen Events

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Brain Storm

We should give some probability mass to those unseen events! This will inevitably reduce the mass for seen events.

- **back-off** to a lower order estimations回退：三元→二元→一元
- produce a pseudo word UNKNOWN
- smooth our raw estimations

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Intuition

If we do not have a higher-order ngram, can we use less context, i.e., related lower-order ngrams ?

Linear Interpolation

Intuition

If we do not have a higher-order ngram, can we use less context, i.e., related lower-order ngrams ?

- Back-off
  - use higher-order estimators if you have enough counts
  - otherwise lower-order, ...
- Interpolation
  - just combine estimators of different orders:

回退实际上是一种线性插值

$$
\begin{aligned}
p_{il}(w_i|w_{i-2}, w_{i-1}) = \quad & \lambda_1 p_{ml}(w_i|w_{i-2}, w_{i-1}) \\
+ \quad & \lambda_2 p_{ml}(w_i|w_{i-1}) \\
+ \quad & \lambda_3 p_{ml}(w_i)
\end{aligned}
$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and $\lambda_i \geq 0$ for all i

  - does it still yield a distribution?

Linear Interpolation

Yes!

Linear Interpolation

Yes! But why?

Held-out Data and Parameter Estimations

How to figure out those λs ?

- hold out part of training data as a validation set
- choose λs to maximize the loglikelihood of the validation set, in the trigram case:

$$\sum_{w_1,w_2,w_3} count_{dev}(w_1, w_2, w_3) \log p_{il}(w_3|w_1, w_2)$$

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- using an iterative approach to estimate λs
- may allow λs to vary according to specific counts

Estimating λs Iteratively

Step 1: Initialization: randomly pick values for λ_1, λ_2, and λ_3
Step 2: Calculating

$$c_1 = \sum_{w_1,w_2,w_3} \frac{count_{dev}(w_1,w_2,w_3)\lambda_1 \cdot p_{ml}(w_3|w_1,w_2)}{\lambda_1 \cdot p_{ml}(w_3|w_1,w_2) + \lambda_2 \cdot p_{ml}(w_3|w_2) + \lambda_3 \cdot p_{ml}(w_3)}$$

$$c_2 = \sum_{w_1,w_2,w_3} \frac{count_{dev}(w_1,w_2,w_3)\lambda_2 \cdot p_{ml}(w_3|w_2)}{\lambda_1 \cdot p_{ml}(w_3|w_1,w_2) + \lambda_2 \cdot p_{ml}(w_3|w_2) + \lambda_3 \cdot p_{ml}(w_3)}$$

$$c_1 = \sum_{w_1,w_2,w_3} \frac{count_{dev}(w_1,w_2,w_3)\lambda_3 \cdot p_{ml}(w_3)}{\lambda_1 \cdot p_{ml}(w_3|w_1,w_2) + \lambda_2 \cdot p_{ml}(w_3|w_2) + \lambda_3 \cdot p_{ml}(w_3)}$$

Step 3: Re-calculate: $\lambda_i = \frac{c_i}{c_1+c_2+c_3}$
Step 4: if $\Delta\lambda_i > \theta$, go to Step 2

Stupid Back-off

- for a fixed vocabulary at a web scale, we have trigrams :

$$s(w_i|w_{i-2}, w_{i-1}) = \begin{cases} \frac{count(w_{i-2},w_{i-1},w_i)}{count(w_{i-2},w_{i-1})} & count(w_{i-2}, w_{i-1}, w_i) > 0 \\ 0.4 s(w_i|w_{i-1}) & \text{otherwise} \end{cases}$$

- Brants et al. 2007 (Google)

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- is this model a probabilistic one?
- NO! $s()$s do not sum to 1

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A Basic Smoothing Solution

Just add **one count** for all possible n-grams!

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Just add **one count** for all possible n-grams!

- Does this work ?
- Any side effect you could imagine ?

Add-One Smoothing

Intuition

For all possible n-grams, just add an extra count.

$$p_{add1}(w_i|w_{i-1}) = \frac{count(w_{i-1}, w_i) + 1}{count(w_{i-1}) + |\mathcal{V}|}$$

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- also called Laplace-smoothing

加一法适用于 大训练集

小训练集→很稀疏→加一片1→被1主导了，不好

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Example

- $|\mathcal{V}| = 12$, $\mathcal{V} = \{\text{horse}, \text{cat}, \text{dog}, \text{dunkey}, \text{the}, ...\}$
- seen: $p(\text{horse}|\text{the}) = 0.5$, $p(\text{cat}|\text{the}) = 0.5$
- 0 for the unseen
- after Add-one
  - previously seen: $p_{add1}(\text{horse}|\text{the}) = 0.18$, $p_{add1}(\text{cat}|\text{the}) = 0.18$
  - previously unseen: $p_{add1}(\text{dog}|\text{the}) = 0.09$, $p_{add1}(\text{dunkey}|\text{the}) = 0.09$, ...

Add-One Smoothing

- $p(\text{SEEN EVENTS}|\text{the})$ from 1 to 0.36
- $p(\text{UNSEEN EVENTS}|the)$ from 0 to 0.64

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Add-One Smoothing

- $p(\text{SEEN EVENTS}|\text{the})$ from 1 to 0.36
- $p(\text{UNSEEN EVENTS}|the)$ from 0 to 0.64
- are we putting too much probability mass on unseen events?
- too many zeros in our example
- suit cases where $\#(\text{UNSEEN})$ is not that huge

Good-Turing Discounting

Intuition: Use the counts of n-grams that are **seen once** to help estimate the counts of **unseen events**

Good-Turing Discounting

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- count of count : N_r, number of words with frequency r
- Translate real counts r into adjusted counts r^*:

$$r^* = (r+1)\frac{N_{r+1}}{N_r}$$

- The probability mass reserved for unseen events is N_1/N_{all}
- For larger r (where N_{r-1} is often 0), various other methods can be applied (curve fitting or linear regression).

Good-Turing Discounting

During testing, how possible we see a word that appears r times in the training set? Imagine we can randomly delete a word from the training set to construct a **new but pseudo** training set:

- everytime, we delete a word w
  - we know w appears $r+1$ times in the original set
  - now it is supposed to appear r times in the new training set
- the total apearances of $r+1$ words in the original set is $N_{r+1}(r+1)$
- after many times of deletion, the number of those words in the new set could be considered as N_r
- the possible/average counts of those words should be $N_{r+1}(r+1)/N_r$

Good-Turing Example

Trained on 22m words of AP News (Church and Gale, 1991)

r	r^*	$r - r^*$
0	0.000027	
1	0.446	0.554
2	1.26	0.74
3	2.24	0.76
4	3.24	0.76
5	4.22	0.78
6	5.19	0.81
7	6.21	0.79
8	7.24	0.76

Absolute Discounting

In Good-Turing discounting,

- It looks like $r^* = r - 0.75$, in most cases

So, we may

- Subtract a fixed number d from each count

$$p_{abd}(w_2|w_1) = \frac{count(w_1, w_2) - d}{count(w_1)} + \lambda(w_1)p(w_2)$$

- Typical counts 1 and 2 are treated differently
- $\lambda(*)$: a weighting function

Kneser-Ney Smoothing

- Popular in the traditional category
- Combines various smoothing method
  - absolute discounting
  - considers diversity of history, predicted words
  - interpolation

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More in LM

- Language modeling is definitely an active field of research
- There are many back-off and interpolation methods
- Skip n-gram models: e.g., back-off to $p(w_n|w_{n-2})$, ...
- Factored language models: back-off to word stems, part-of-speech tags, ...
- Syntactic language models: using parse trees

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- Factored language models: back-off to word stems, part-of-speech tags, ...
- Syntactic language models: using parse trees
- Language models trained on billions and trillions of words

Pros and Cons

- Really easy to build, on billions of words
- Smoothing helps generalize to new data
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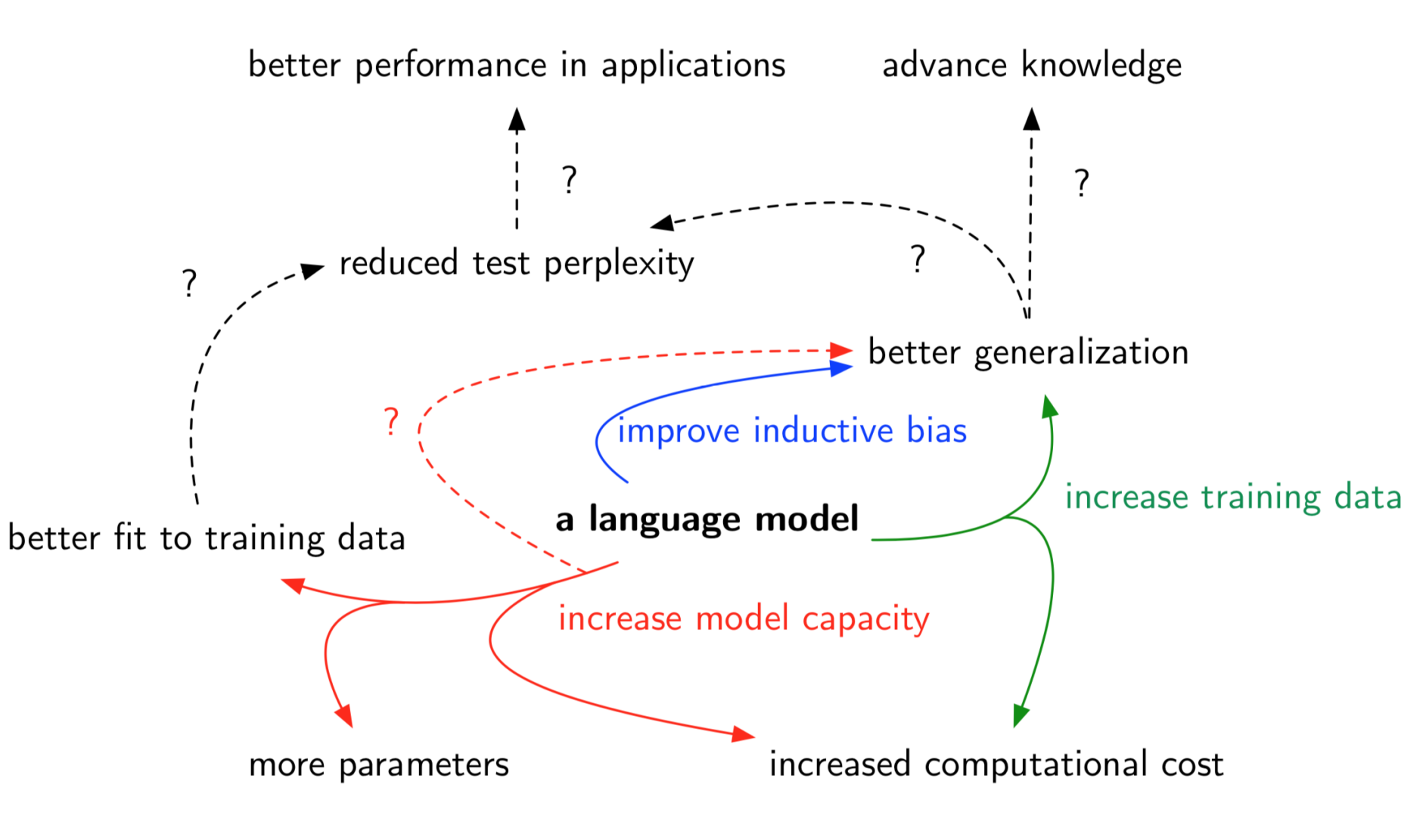

[from Noah Smith]

The Magic

modeling the context

The Magic

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- history: 1-4 words
- history: 3-5-7 words window

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Seems that we are always counting words

Now

Questions

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- can we go beyond *counting words*?
- can we use something else learned before?
- can we train on *much more words*?

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Questions

- can we go beyond *counting words*?
- can we use something else learned before? *classification*?
- can we train on *much more words*?

Readings

- Chapter 3, Speech and Language Processing (3rd. **SLP**)
- SRI Language Modeling Toolkit, http://www.speech.sri.com/projects/srilm/manpages/
- Improved backing-off for m-gram language modeling, Reinhard Kneser and Hermann Ney, ICASSP 1995.
- A comparison of the enhanced Good-Turing and deleted estimation methods for estimating probabilities of English bigrams, Kenneth W.Church and William A.Gale, Computer Speech & Language, 1991, V5-1, pp19-54
- Large Language Models in Machine Translation, Thorsten Brants, Ashok C. Popat, Peng Xu, Franz J. Och, Jeffrey Dean, EMNLP 2007
- Google N-grams: https://blog.research.google/2006/08/all-our-n-gram-are-belong-to-you.html
- Google Book N-grams: https://books.google.com/ngrams/